

Colorectal Cancer: Comprehensive Management

Screening, Molecular Subtypes, Surgery, and Systemic Therapy

Doc #09 of 15	Domain: Oncology & Cancer Treatment	Audience: Medical Professionals
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1. Epidemiology & Screening

CRC is the 3rd most common cancer worldwide (~1.9 million new cases/year) and 2nd leading cause of cancer death. Incidence peaks at ages 60-75; ~5% attributable to Lynch syndrome (germline MMR gene mutations: MLH1, MSH2, MSH6, PMS2). Screening modalities approved by major guidelines:

Modality	Interval	Sensitivity (CRC)	Notes
Annual FIT (faecal immunochemical test)	1 year	~79%	Non-invasive; preferred in average-risk
Multi-target stool DNA (Cologuard)	3 years	~92%	Also detects advanced adenomas
Flexible sigmoidoscopy	5 years	~60-70% (left colon only)	Misses proximal lesions
Colonoscopy	10 years	~95%	Gold standard; allows polypectomy
CT colonography	5 years	~96% (≥10mm polyps)	No biopsy capability; radiation

2. Molecular Pathways & Consensus Molecular Subtypes (CMS)

CMS	% Cases	Molecular Features	Prognosis
CMS1 (MSI immune)	14%	MSI-H, BRAF V600E, hypermutation, immune infiltration	Good OS; poor post-relapse survival
CMS2 (canonical)	37%	WNT/MYC activation, CIN, RAS/EGFR downstream	Good prognosis, best post-relapse
CMS3 (metabolic)	13%	KRAS mutation, IGFBP2, mixed MSI/CIN	Intermediate
CMS4 (mesenchymal)	23%	TGF- β activation, stromal infiltration, worst prognosis	Poor, especially relapse survival

3. Adjuvant Therapy — Stage II-III

- Stage III (node+): 6 months FOLFOX (or CAPOX 3-4 months per IDEA trial) standard of care; 3-month CAPOX acceptable for low-risk T1-3N1
- Stage II high-risk (T4, perforation, obstruction, <12 nodes sampled, pN0 after inadequate dissection): Consider FOLFOX/CAPOX; 5-FU/capecitabine alone if lower risk
- Stage II MSI-H: Adjuvant chemotherapy has minimal benefit (disproportionately good prognosis); observation preferred
- Anti-EGFR agents (cetuximab/panitumumab) are NOT beneficial in the adjuvant setting (NCCTG N0147, PETACC8)

4. Metastatic CRC — First-Line Treatment Algorithm

Molecular stratification is essential before initiating treatment:

- MSI-H / dMMR (5% of mCRC): Pembrolizumab 1st line (KEYNOTE-177 — superior PFS/OS vs. chemotherapy)
- KRAS/NRAS/BRAF wild-type (left-sided primary): FOLFOX or FOLFIRI + cetuximab or panitumumab (RAS/RAF wt required for anti-EGFR)
- KRAS/NRAS mutant: FOLFOX or FOLFIRI + bevacizumab; FOLFOXIRI + bevacizumab for fit patients with high volume disease
- BRAF V600E mutant: FOLFOXIRI + bevacizumab (1L); encorafenib + cetuximab (2L; BEACON-CRC2)
- HER2 amplified (2-3% of RAS/BRAF-wt mCRC): Trastuzumab + pertuzumab or T-DXd (2L+)

CLINICAL PEARL: Primary tumour sidedness (left vs. right) is an independent predictive biomarker for anti-EGFR response in RAS-wt mCRC. Left-sided tumours derive significant OS benefit; right-sided tumours do not — use bevacizumab-based regimens instead.